

			(prop x n)	
(i)	Classification	No of Std	proportion	Sample size
	B.Sc	150	150/728	$\frac{150}{728} \times 40 = 8$
	B.A	163	163/728	$\frac{163}{728} \times 40 = 9$
	F.Sc	195	195/728	$\frac{195}{728} \times 40 = 11$
	F.A	220	220/728	$\frac{220}{728} \times 40 = 12$
		728		40

Final results

The sample size for each classification are

B.Sc : 8 students

B.A : 9 students

F.Sc : 11 Students

F.A : 12 students

A.B

(ii) Given:

$$n = 100$$

$$\hat{p} = \text{Number of success} = 16 = 0.2133$$

$$\text{Total sample size } 75$$

$$Z = 1.645 \text{ (from } Z \text{ table for 90\% confidence interval)}$$

Calculate standard error

$$S.E. = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.2133(1-0.2133)}{75}}$$

$$= \sqrt{\frac{0.2133 \times 0.7867}{75}} = 0.002236$$

$$= 0.0473$$

$$M.E. (\text{margin of error}) = Z \times S.E. = 1.645 \times 0.0473 = 0.0778$$

$$C.I. = \hat{p} \pm M.E. = 0.2133 \pm 0.0778$$

$$C.I. = 0.2133 + 0.0778 \quad C.I. = 0.2133 - 0.0778$$

$$C.I. = 0.2911$$

$$C.I. = 0.1355$$

(iii) Step 1: (Test hypothesis):

$$H_0: \mu \geq 50$$

$$H_a: \mu < 50$$

Since sample size is 36 (large enough)
we use z test for test statistics

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Here $\bar{x} = 48.7$

$$\mu_0 = 50$$

$$s = 3.1$$

$$n = 36$$

Putting values

$$z = \frac{48.7 - 50}{3.1/\sqrt{36}} = -1.3$$

$$z = -1.3$$

$$0.5167$$

$$z = -2.52$$

Here $\alpha = 0.05$ (typical threshold for
rejection ~~for~~ in hypothesis test)

$$z_{\text{critical } (0.05)/2} = 0.975$$

$$z_{\text{critical}} = 1.645$$

The calculated z-value is -2.52

critical value is -1.645

$$-2.52 < -1.645$$

The test hypothesis falls for in the rejection region.

Reject the null hypothesis, there is a significant evidence to conclude that average age of (student) members of parliament is less than 50 years.

(iv) Step 1:-

$$H_0: \mu_2 - \mu_1 \leq 10$$

$$H_a: \mu_2 - \mu_1 > 10$$

Given:-

$$n_1 = 12, \bar{x}_1 = 10, s_1^2 = 1200$$

$$s_1 = \sqrt{1200} = 34.64$$

$$n_2 = 18, \bar{x}_2 = 25, s_2^2 = 900$$

$$s_2 = \sqrt{900} = 30, \Delta_0 = 10$$

Significance level $= \alpha = 0.05$

$$t = \frac{(\bar{x}_2 - \bar{x}_1) - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$l = (15 - 10)$$

$$\sqrt{\frac{1200}{15} + \frac{900}{180}}$$

$$= \frac{5}{\sqrt{100 + 50}} = \frac{5}{\sqrt{150}} = 12.247$$

$$t = 0.408$$

Determine the degree of freedom

$$df = \left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2$$

$$= \left(\frac{s_1^2}{n_1} \right)^2 + \left(\frac{s_2^2}{n_2} \right)^2$$

$$= \left(\frac{1200}{12} + \frac{900}{18} \right)^2$$

$$= \left(\frac{1200}{12} \right)^2 + \left(\frac{900}{18} \right)^2$$

$$= \frac{(100 + 50)^2}{11 + 17}$$

$$= \frac{22500}{909.09 + 147.06}$$

$$df = \frac{22500}{1056.15} = 21.3$$

$$df = 21$$

• for critical angle

$$\alpha = 0.05 \text{ and } df = 21$$

$$t_{\text{critical}} = 1.721$$

(0.05)(2)

• Compare t value with t critical

$$t = 0.408 < 1.721$$

we fail to reject the null hypothesis

Therefore $\alpha = 0.05$ level to support the claim that

$$\mu_2 - \mu_1 > 10$$

(V) Step 1. **PUACP**

$$H_0: \sigma^2 \leq 7.7953$$

$$H_a: \sigma^2 > 7.7953$$

Sample (x)	$(x - \bar{x})$	$(x - \bar{x})^2$
67.50	-0.15	0.025
70.75	3.1	9.6225
72.00	4.35	18.9225
63.25	-4.4	19.6025

11/12

x	$(x - \bar{x})$	$(x - \bar{x})^2$
65.25	-2.4	5.6225
68.75	1.1	1.1025
69.50	1.85	2.7225
68.50	0.85	0.7225
66.50	-1.15	1.3225
64.75	-2.9	8.4025
676.50		68.36

$$\bar{x} = \frac{\sum x}{n} = \frac{676.50}{10}$$

$$\bar{x} = 67.65$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{68.36}{10-1}$$

$$s^2 = \frac{68.36}{9} = 7.596$$

The test statistics for variance uses the chi-square distribution.

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

$$= \frac{(10-1)(7.596)}{7.7953} = \frac{9 \times 7.596}{7.7953}$$

$$= \frac{68.364}{7.7953}$$

$$\chi^2_{(0.01), (9)} = 21.666$$

$$\text{Test statistic } (\chi^2) = 8.77$$

$$\text{Critical value} = 21.666$$

$$\chi^2 < \chi^2_{\text{critical}}$$

$$8.77 < 21.666$$

we fail to reject the null hypothesis there is insufficient evidence at the $\alpha = 0.01$ level to conclude that $\sigma^2 \neq 7.7953$

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PUACP